Retail Sales & Business Performance Analytics

Week 6 — Git, GitHub & Final Data Analytics Capstone

Prepared as a reproducible educational project package

Dataset disclosure: The included dataset is synthetic demonstration data with intentionally inserted data-quality issues. It is not real company data. If the evaluator requires a real-world dataset, replace it with a cited public dataset and rerun the analysis before submission.

1. Project Overview

This project demonstrates an end-to-end retail analytics workflow: inspecting raw data, cleaning records, performing exploratory data analysis, creating visualizations, extracting business insights, and preparing an implementation guide for an interactive Power BI dashboard.

2. Problem Statement

Retail decision-makers need to understand sales, profit, order volume, product/category mix, customer segments, and regional performance. The project provides a repeatable way to transform transaction-level records into metrics that support inventory, pricing, and campaign decisions.

3. Dataset Description

The raw CSV contains 1,203 rows including injected missing values, invalid values, and duplicates. After cleaning, 1,190 rows remain. The date range is 2024-01-01 through 2025-12-30. Monetary amounts are generic currency units because no currency was specified.

Fields: Order_ID, Order_Date, Region, Category, Product, Quantity, Unit_Price, Discount, Sales, Profit, Customer_Segment, Payment_Method.

4. Data Cleaning Process

• Parsed date and numeric columns with explicit types.

• Removed duplicate records and rows missing critical order/date/category/product identifiers.

• Removed non-positive quantities, negative unit prices, and discounts outside 0–100%.

• Imputed categorical nulls with the mode and missing unit prices with the median.

• Recalculated Sales = Unit_Price × Quantity × (1 − Discount).

• Filled missing Profit using the category median profit margin, with global fallback.

• Saved the resulting data as data/cleaned_sales.csv.

5. Exploratory Data Analysis

The notebook includes shape, data types, missing-value counts, duplicate counts, descriptive statistics, category summaries, numeric correlations, and IQR-based outlier screening. IQR flags are a review aid, not automatic proof of errors. Correlation does not imply causation.

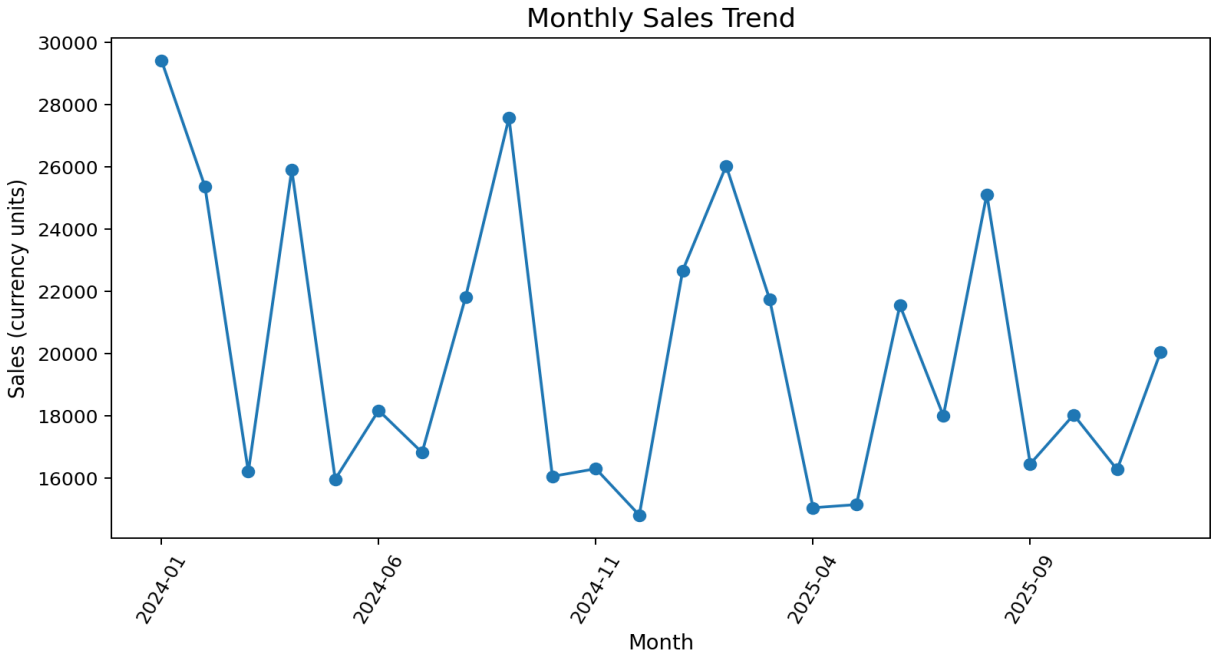
6. KPI Summary

Metric	Value
Total Sales	480,580.26
Total Profit	178,868.70
Distinct Orders	1,190
Units Sold	3,194
Profit Margin	37.22%
Average Order Value	403.85

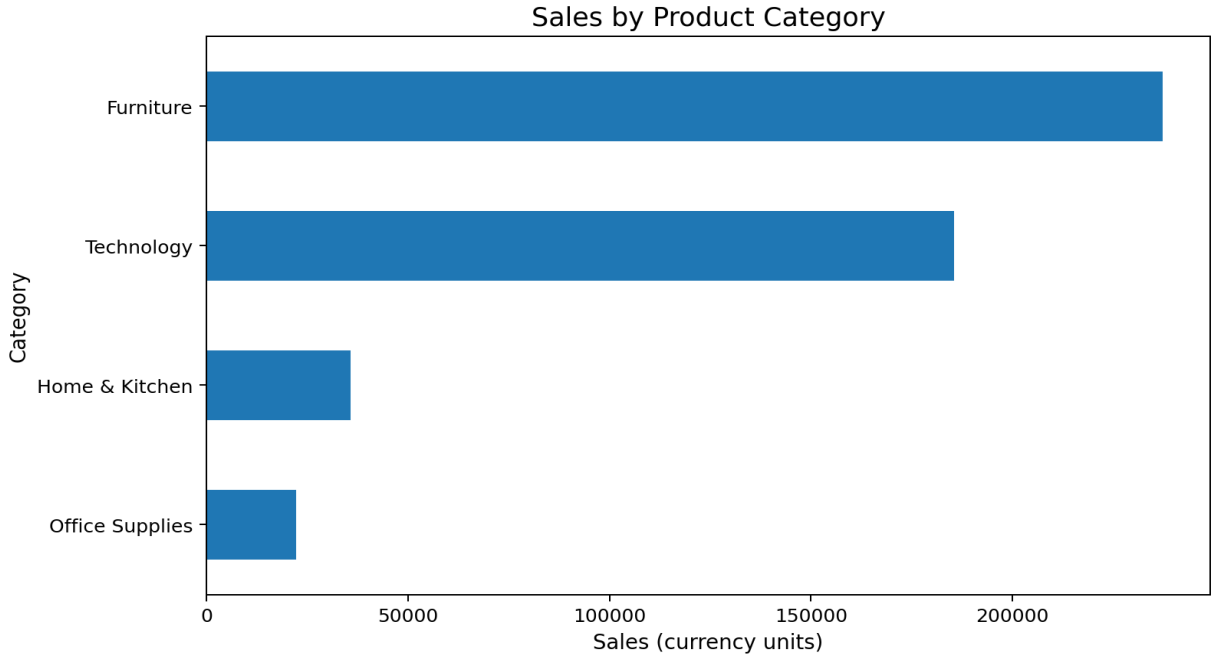
These figures describe only the synthetic dataset supplied with this package.

7. Visualizations

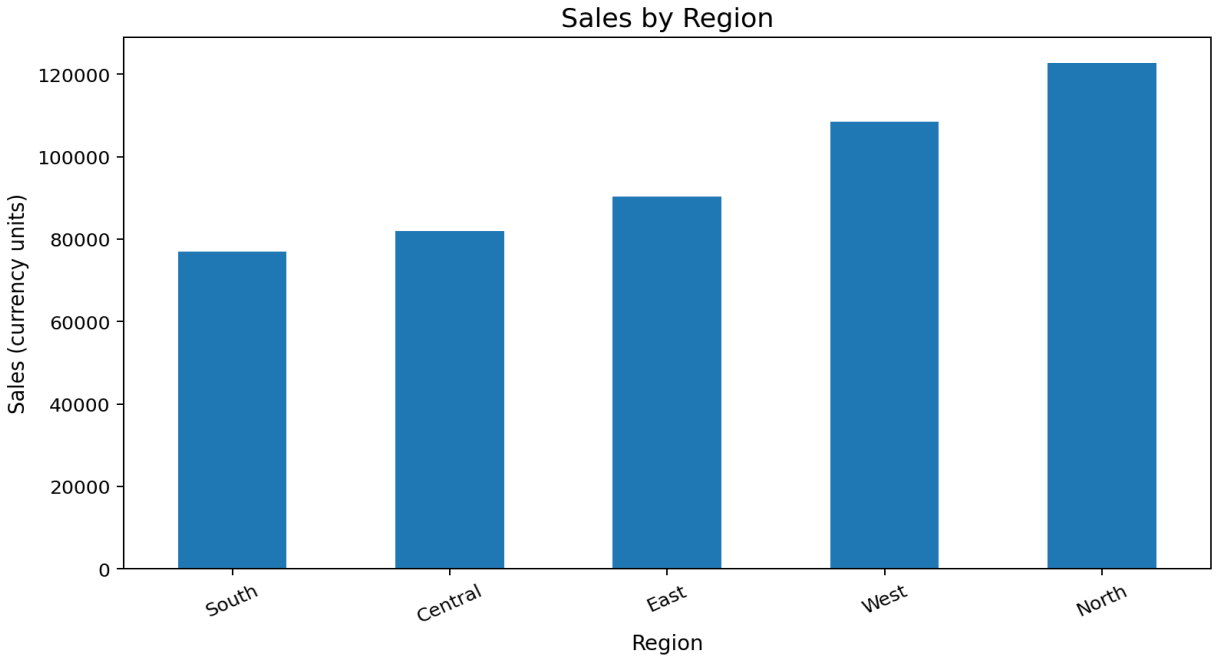
Monthly Sales



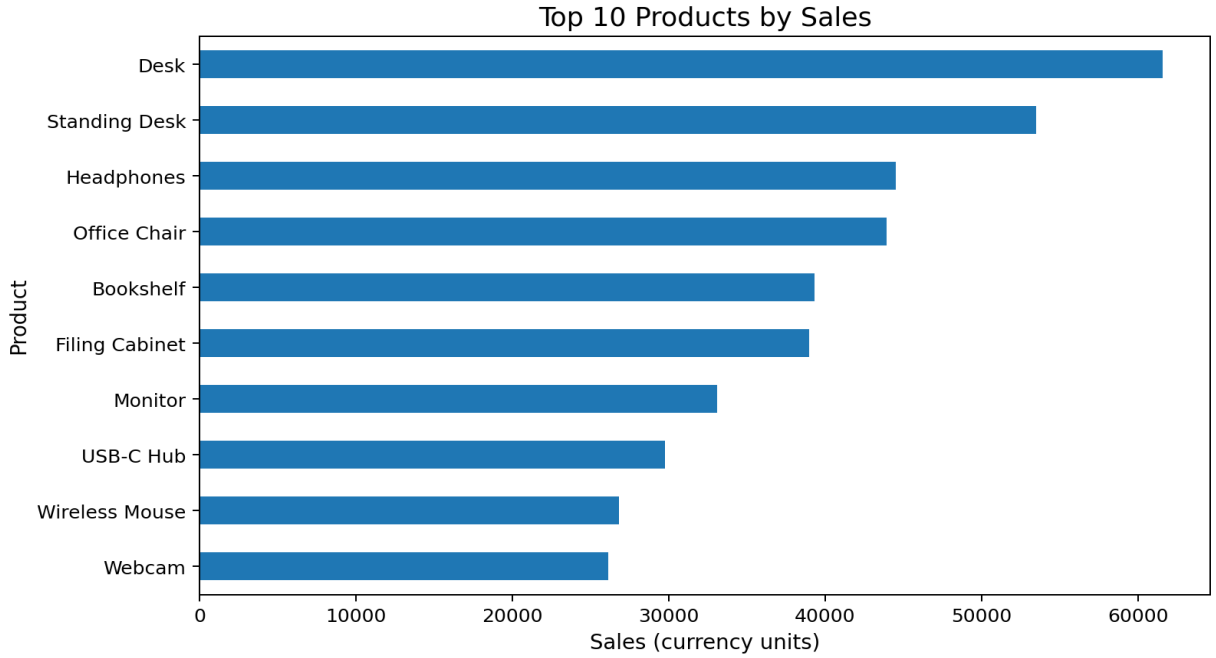
Category Sales



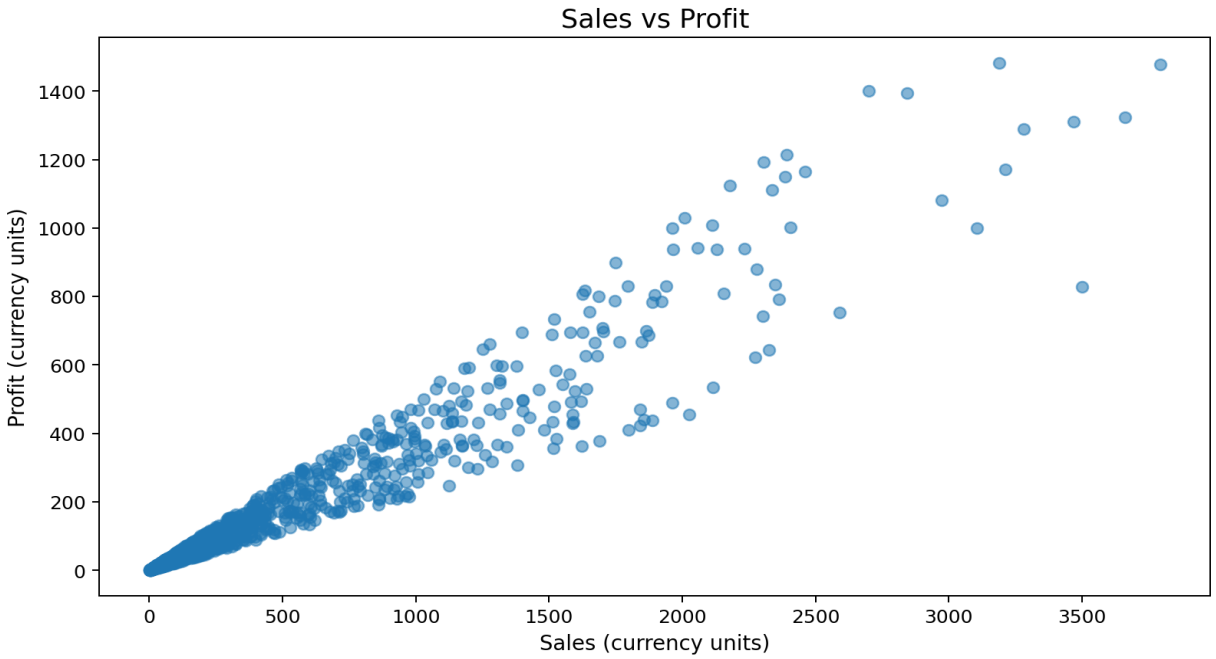
Regional Sales



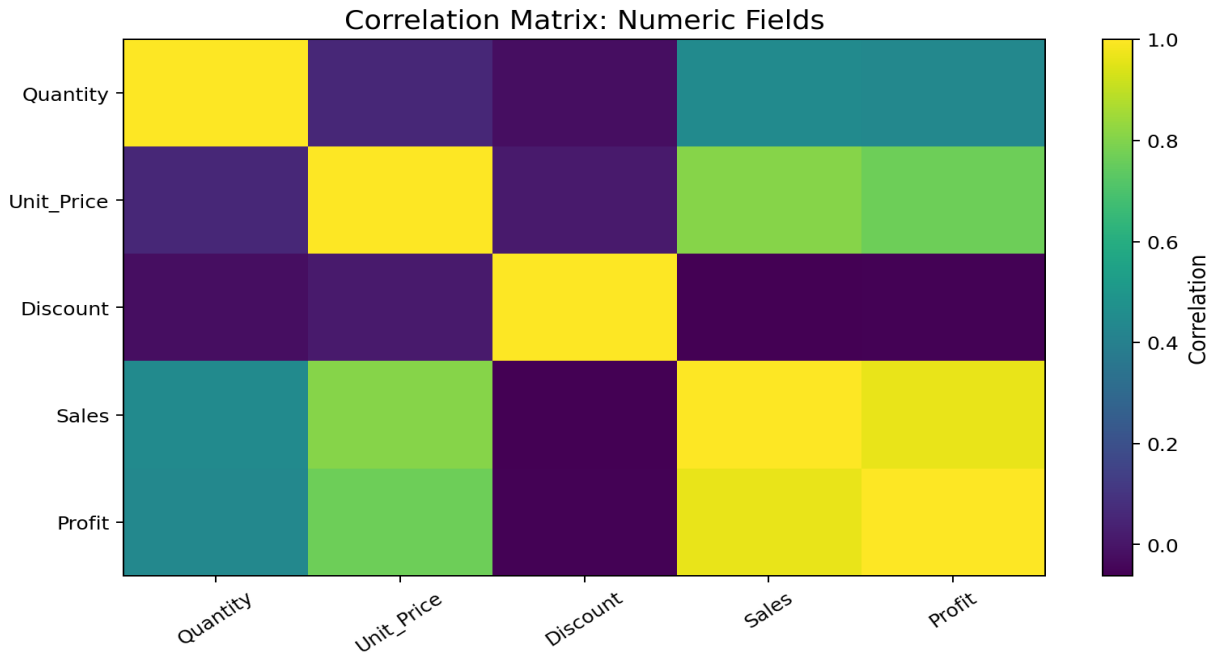
Top Products



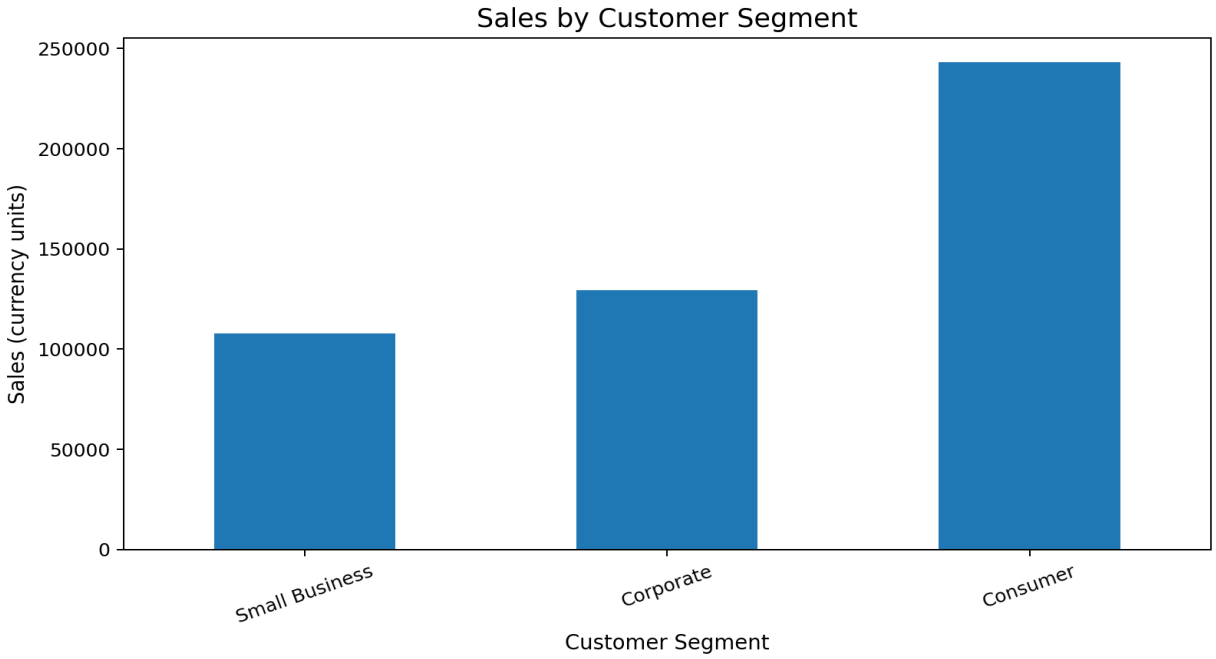
Sales Profit Scatter



Correlation Heatmap



Segment Sales



8. Business Insights

• Total sales are 480,580.26 across 1,190 distinct orders.

• Aggregate profit margin is 37.22%; sales should be assessed alongside profitability.

• Furniture is the highest-sales category in this sample.

• North is the highest-sales region in this sample.

• Desk is the highest-sales product in this sample.

• Office Supplies has the highest aggregate category margin; Technology has the lowest.

• 2024-01 is the highest-sales month and 2024-12 the lowest in this generated sample.

9. Recommendations

• Use monthly and category trends to inform replenishment and campaigns, validated against actual stockouts, inventory, and lead-time data.

• Track discounting alongside profit margin and run controlled promotion tests focused on incremental profit, not revenue alone.

• Review product performance together with margin, returns, and inventory holding cost before changing the assortment.

• Normalize regional performance by stores, customers, and marketing spend where available.

10. Power BI Dashboard Plan

The separate Power BI guide specifies four KPI cards, a monthly sales line chart, category sales/profit comparison, regional sales, top products, customer segment mix, and slicers for date, region, category, segment, and payment method. DAX measures are provided.

Build status: A compiled .pbix file is not included. It must be created and saved in Power BI Desktop using the supplied cleaned CSV and instructions.

11. GitHub Workflow

The repository includes exact Git setup, clone, add, commit, push, pull, and verification commands. The suggested workflow contains three meaningful commits.

12. Conclusion and Limitations

The project demonstrates a reproducible analytics workflow from raw data through business recommendations. The included dataset is synthetic, so findings are illustrative only. For a real-world submission, cite the public dataset source, replace the data files, rerun the notebook and script, regenerate charts, and update report/dashboard conclusions.